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Mobile computing
IT405089

CIA-1

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(1) Given

Hexagonal cellular network region = 1200km²

Radius = 1.8km

Band width = 48MHz & 200MHz

cluster size N=7 & N=4

Reduction = 15%.

No. of cells = ?

Total available channels:

Reuse ratio $Q = 4.5$

$Area = \pi R^2$

$= 1200(1.8)^2$

$= 1200 (1.8 \times 1.8)$

$= 1200 (3.24)$

Area : 3888800

No. of cells = Area * Bandwidth

$= 3888 \times 200$

$= 777600$

Available channels = cluster size * Reduction

$= 7 \times \frac{15}{100}$

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$= \frac{7 \times 15}{100}$

$= \frac{700 \times 15}{100} \Rightarrow \frac{10500}{100}$

Avaiable channels = 105.

Ratio = 4.5

Radius = 1.8 km

$= \frac{4.5 \times 18}{1}$

$= 8.10$

∴ Suitable confiration will be = 7//

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(2)

Given

Time = 40 mins

No. of cells = 15 with 90 km/h

PLMN = 2

handoff = 120ms coordination 80ms

PLMN additional 350 & 250 ms.

3 fail additional 400ms

Total normal handoff delay = ?

Recovery delay = ?

= 512 kbps

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No. of cells = PLMN x MS
= 2 x 80MS
= 160

Limit of 3 seconds.

Normal hand off delay = ?

$$= \frac{\text{time} \times \text{cells}}{\text{hand off}}$$

$$= \frac{40 \times 15}{120}$$

$$= \frac{600}{120}$$

$$= 5.4$$

Normal hand off delay = 5.4

Recovery delay = ?

No. of cells/speed

$$= 90/15 \Rightarrow 6$$

2 inter PLMN = $6/2 = 3$

Recovery delay = 3.

∴ Normal handoff delay - Recovery delay

$= 5.4 - 3$

Improvement to Bss $= 2.4$

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(3) i) Given

Base Transceiver stations = 12

Sectors = 3

TRXs = 4

TDMA provided = 8 which 2 tym slots.

calls per hour = 1.2

E1 link of = 2048 Mbps 80%.

No. of TRX's required per Sector =

= time * sectors

= 2 x 3 = 6 sectors.

E1 Link of = 2.048 MBPS

percentage = $\frac{12 \times 80}{100} = \frac{960}{100} \Rightarrow \frac{1200 \times 1.2 \times 96}{3600}$

= 96%.

$= 360$

Percentage = 96%.

Trxs =

1 → 3

8 → 1 = 12

Trx's = time * percentage

= 12 * 96

= 1152

per each Sector = 2

= 1152 for TRX's

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(6)

each lasting 0.5 sec.

Time = 384kbps

12 hour emergency data delivery

D = 240 × 18 = 4320 MB

DT = 4320 (1.112) = (4838.4)

= 4838.4 × 8 = 38707.2 MD

Data rate =

R = 4 × 20

= 80 kbps

Protocol = 80 (1 − 0.18) = 6.5.kbs.

= 65 × 6 × 05/100 = 25.75 kbps

$$T = \frac{38702.2}{0.384} = 100800 \text{ kbps}$$

T = 28 kBPS

$$\text{Reduction} = \frac{192.91 - 28}{192.91} \times 100$$

= 85.494

Conclusion:-

GPRS fails on 12 hours deadline. UMTS also fails But UMTS is faster than GPRS. So UMTS is faster.

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⑤

Given

120 GB Software

Mobile devices = 12000

GSM provides = 9.6kbps

throughput = 64

Percentage = 10% 40% transfer data

∴ 1GB = 1000MB , 1MB = 8 megabits

produced = 120000/2000 = 60MB

= 60 × 8 = 480MB

GPRS = 40% of devices transfer data

$= \frac{12000 \times 40}{100} = 800$

1GB = 1000MB

Devices (5G)

800 × 2 MBPs

85 × 60 mbps/min

= 1500 mPs/min.

Per hour = 1500 × 60

9000 MBPS/h

1GB = 1000 MB

per hour = 90GB

∴ Total time taking for 5G devices to update their devices = 1hour 20 mins.

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= 1152×2

= 2304

Conclusion

∴ El Abis link remains unsufficient after the upgrade.

ii) Given

oneway 10MHz

N = 7

frequency = 30

factor = 0.4 of 0.6

Spectral efficiency & interference

chip rate = 1.2288 Mbps to 9.6 Kbps

N = 1

frequency = 30 × 0.4

= 1.20

= 1.2288 MbPS

upgrade to 5G from 3G.

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(4) Given

Network transfers = 240

size = 18MB

Theoretical data = 20 kbps

Reduce capacity = 18%

throughput 85% & 12%.

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